

Day 29 – Iterator Design Pattern

Introduction

👉 Iterator Design Pattern is used to **traverse (iterate) elements of a collection**.

- Har data structure me traversal hota hai
- Traversal depends on:
 - Linked List → pointer traversal
 - Binary Tree → preorder, inorder, postorder
 - Vector → loop traversal

Problem

👉 Different DS = different traversal logic

```
Vector → for loop  
LinkedList → temp pointer  
Tree → recursion
```

👉 Yeh sab client ko pata hona chahiye ❌ (bad design)

Solution → Iterator

👉 Common interface:

```
hasNext()  
next()
```

- hasNext() → check karta hai ki next element available hai ya nahi
- next() → next element return karta hai aur pointer aage move karta hai

Short:

hasNext = check

next = value + move

👉 Client ko DS ka internal structure nahi pata hona chahiye

Why Iterator Needed?

Without Iterator

```
Playlist (vector)
→ for loop
```

```
Playlist (linked list)
→ pointer traversal
```

- Initially hum vector mein store kar rahe the songs ko so we need to iterate using for loop - for(songs :song)
- Later, we want to store in linked list so we need the traversing logic as it is different for Linked List.

👉 Agar DS change ho gaya: → code bhi change karna padega 

Problem

👉 SRP (Single Responsibility Principle) break ho raha hai becoz we storing business logic and traversing logic in same class.

```
Playlist:
✓ business logic
✗ traversal logic bhi
```

With Iterator

👉 Playlist ko traversal ka idea nahi hoga

```
Playlist → iterator.getNext()
```

👉 Clean separation:

```
Business Logic ≠ Traversal Logic
```

- which means playlist should not know how to traverse , it just call one method and get another song

Key Idea

👉 Iterator = traversal logic ko alag class me daal do

Playlist → sirf call karega
 Iterator → traversal karega

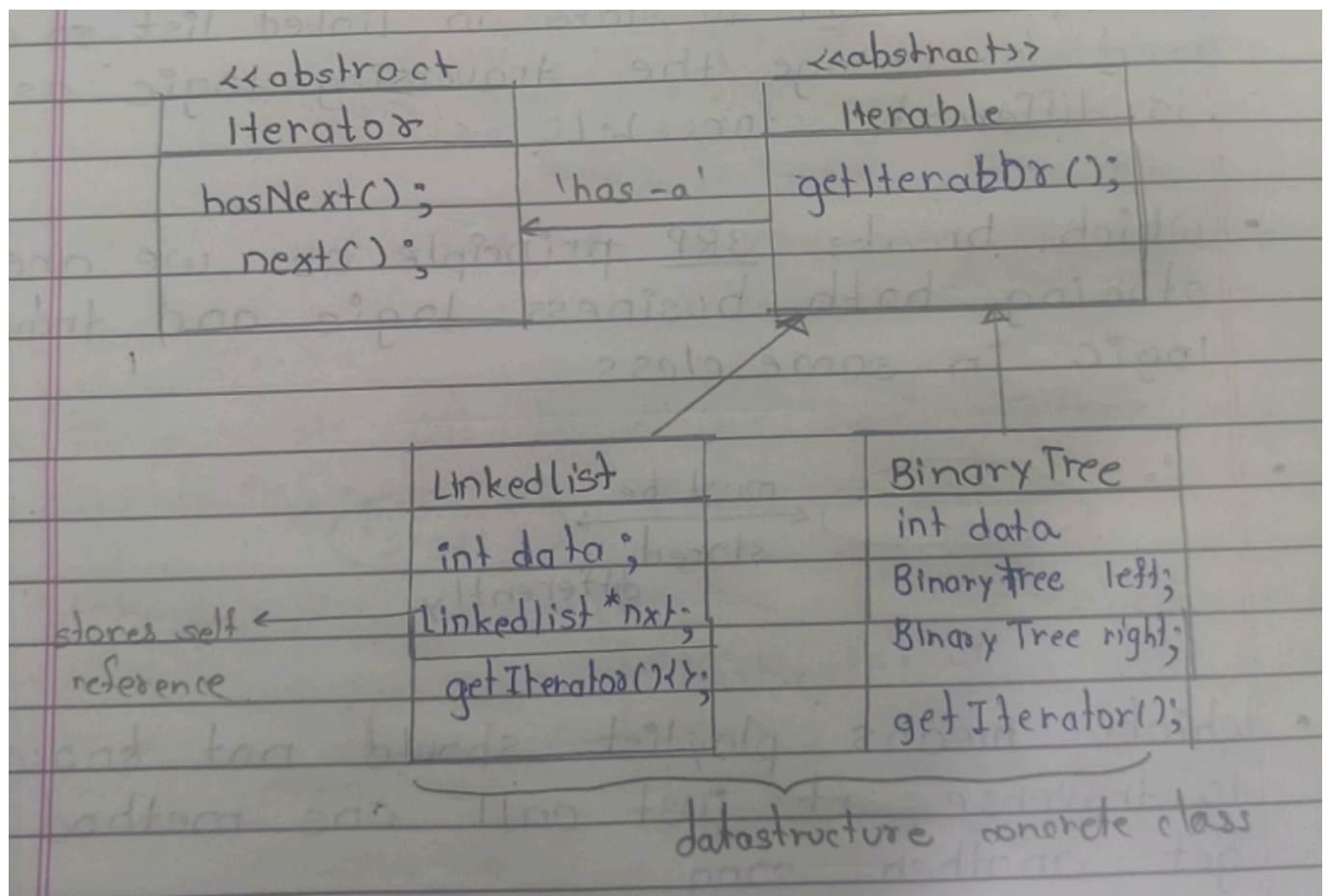
- Methods par koi effect hi nhi aayega becoz hum changes nhi krege as they will just call hasNext() and next() method of iterator passed to them.

🔥 Benefit

✓ SRP follow ✓ DS change hone par code safe ✓ Flexible

■ UML Diagram of Example

Concrete Data Structure

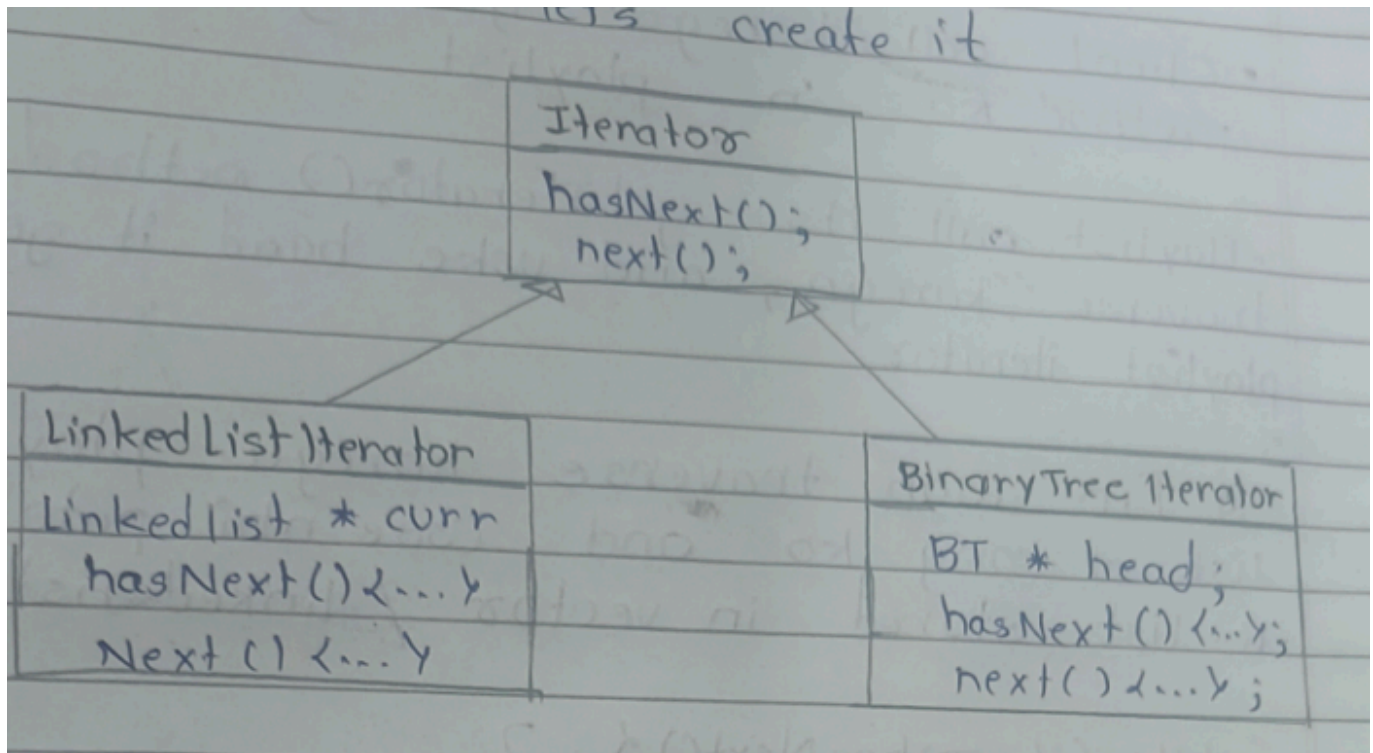


- Ab humne han DS mein baar baar getIterator() method banane ke jagah abatract class bana diya h named as Iterable. (any DS which can be iterable should inherit it)

💬 Meaning

👉 Iterator → traversal ka logic 👉 Iterable → jo traverse ho sakta hai

Concrete Iterators



- `getIterator()` method of every DS hamko return karega ek iterator.

🔍 Traversal Logic (LL)

```

hasNext():
    if curr->next != NULL → true
    else false

next():
    curr = curr->next
  
```

🟢 Important Relation

```

LinkedList "has-a" LinkedListIterator
BinaryTree "has-a" BinaryTreeIterator
  
```

👉 Each DS has its own iterator

🟢 Playlist Example Flow

- Client call karega `getSongByName()` method ko in playlist.
- Playlist call its `getIterator()` method joh traverse karega, and uske baad it gets its playlist iterator.

- Fir woh traverse karega playlist ke list of song ko and usse nhi pata hoga ki list stored in vector / linked list, etc.

Step-by-step Flow

```
Client → playlist.getSongByName()
```

```
Playlist:  
  → getIterator()
```

```
Iterator:  
  while(hasNext()):  
    next()
```

👉 Until required song found asked by client.

🔥 Key Insight

👉 Client ko nahi pata:

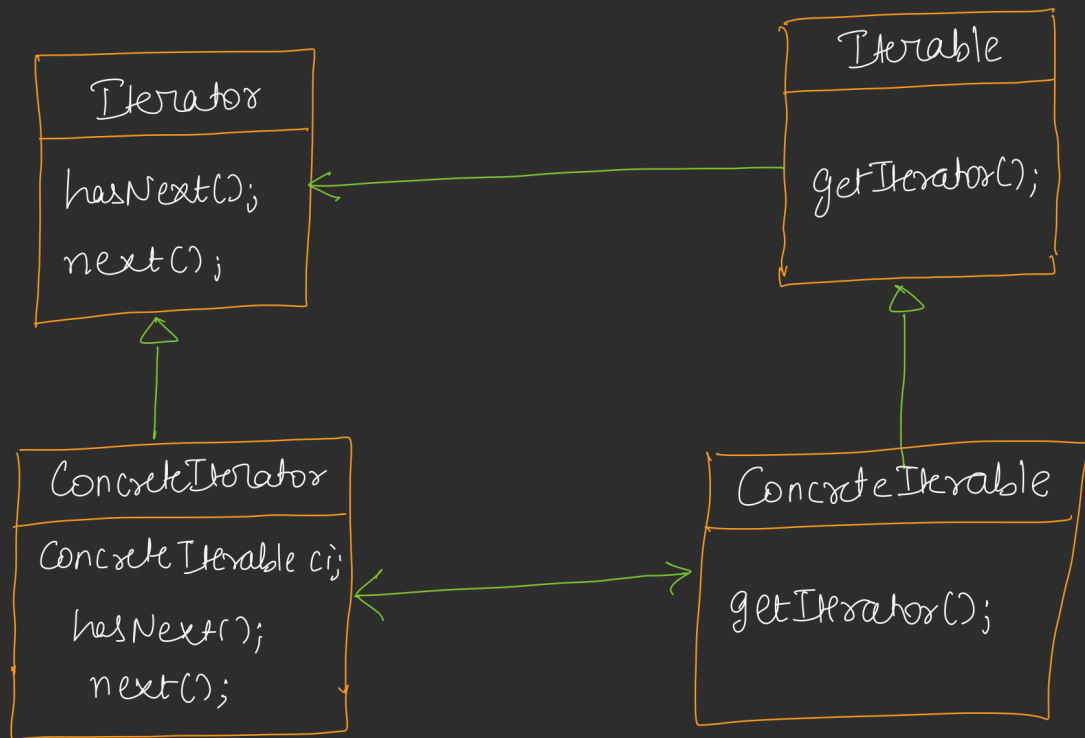
- vector hai
- linked list hai

👉 Sirf iterator use karega

Final UML of Example



Standard UML



Standard Definition

👉 Iterator provides a way to access elements of a collection sequentially without exposing its internal structure.

Real Life Example

👉 Netflix Playlist / Spotify Playlist

Songs stored in:

- ✓ Array
- ✓ LinkedList
- ✓ DB

User:

just presses "Next"

👉 User ko internal structure nahi pata



Advantages

- ✓ SRP follow
 - ✓ Clean code
 - ✓ Flexible traversal
 - ✓ Easy to extend
 - ✓ Data structure change safe
-



Disadvantages

- ✗ Extra classes
 - ✗ Slight complexity increase
-



Interview One-Line

- 👉 Iterator hides traversal logic and gives a standard way to iterate over any collection.
-



Super Simple Summary



Without Iterator:

Client → knows DS → BAD

With Iterator:

Client → only uses hasNext(), next() → GOOD